

DTSA 5798

Supervised Text Classification for Marketing Analytics

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	AI Engineer (67.5 / 100)	57.5 / 100
SPECIALIZATION Marketing Analytics		

58

CURRENT AVG SCORE
across 6 job roles

95

POTENTIAL SCORE
with recommended additions

+50

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Marketing data often requires categorization, or labeling. In today’s age, marketing data can also be very big, or larger than what humans can reasonably tackle. In this course students will learn how to use supervised deep learning to train algorithms to tackle text classification tasks. Students will walk through a conceptual overview of supervised machine learning, and dive into real-world datasets through instructor-led tutorials in Python. The course will conclude with a major project.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	67.5	100.0	+32.5	HIGH
ML Engineer	64.9	95.0	+30.0	HIGH
Machine Learning Engineer	62.5	97.0	+34.5	HIGH
Data Scientist	56.7	96.0	+39.3	HIGH
Applied Scientist	50.5	100.0	+49.5	HIGH
MLOps Engineer	42.9	81.5	+38.6	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Business Communication 2 of 6 roles

insights

business

LLMs & Generative AI 2 of 6 roles

storage

Deep Learning 1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **MEDIUM** **Consider Business Communication content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.

3 **MEDIUM** **Consider LLMs & Generative AI content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: storage.

4 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

AI Engineer

67.5 → 100.0 (+32.5 pts)

following

models

build

technical

systems

problems

technology

storage

ML Engineer

64.9 → 95.0 (+30.0 pts)

discovery

systems

models

world

build

problems

impact

new

Machine Learning Engineer

62.5 → 97.0 (+34.5 pts)

sets

analytics

tools

business

mobile

design

science

models

Data Scientist

56.7 → 96.0 (+39.3 pts)

sets

analytics

mitigation

insights

different

social

build

problems

Applied Scientist

50.5 → 100.0 (+49.5 pts)

discovery

scientific

research

storage

literature

problems

models

technical

MLOps Engineer

42.9 → 81.5 (+38.6 pts)

detection

systems

image

training

tools

models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.